

CURRENT ELECTRICITY

LEVEL 2 – BASIC MCQs (200 Questions)

SECTION 1: ELECTRIC CURRENT & CHARGE FLOW (Q1–30)

1. **Current is defined as:**
 - A. Charge per unit time
 - B. Voltage per unit resistance
 - C. Energy per unit time
 - D. Power per unit voltage
2. **SI unit of electric current is:**
 - A. Coulomb
 - B. Ampere
 - C. Ohm
 - D. Volt
3. **Charge on one electron is:**
 - A. $1.6 \times 10^{-19} \text{ C}$
 - B. $3.2 \times 10^{-19} \text{ C}$
 - C. $6.4 \times 10^{-19} \text{ C}$
 - D. 1 C
4. **How many electrons pass through a cross-section if 1 C of charge flows?**
 - A. 6.25×10^{18}
 - B. 6.25×10^{18}
 - C. 3.12×10^{18}
 - D. 1.6×10^{19}
5. **Conventional current flows from:**
 - A. Negative to positive
 - B. Positive to negative
 - C. Low to high potential
 - D. Both A and C
6. **In a conductor, current is due to:**
 - A. Positive charge motion
 - B. Negative charge motion
 - C. Both A and B
 - D. Fixed electrons

7. **DC stands for:**
- A. Direct Charge
 - B. Direct Conductor
 - C. Direct Current
 - D. Direct Conductivity
8. **If current $I = 5\text{ A}$ and time $t = 2\text{ s}$, charge Q is:**
- A. 2.5 C
 - B. 10 C
 - C. 7 C
 - D. 5 C
9. **In a circuit, if charge flows at rate 10 C/s , current is:**
- A. 5 A
 - B. 10 A
 - C. 20 A
 - D. 100 A
10. **Current density J is defined as:**
- A. Current per unit area
 - B. Charge per unit volume
 - C. Current per unit length
 - D. Voltage per unit area
11. **If current through a wire is 2 A , charge in 5 seconds is:**
- A. 2.5 C
 - B. 5 C
 - C. 10 C
 - D. 20 C
12. **Electrons in a conductor move:**
- A. With constant velocity
 - B. With accelerating drift velocity
 - C. With random thermal motion + drift
 - D. Without any motion
13. **Direction of electron flow is:**
- A. Same as conventional current
 - B. Opposite to conventional current
 - C. Perpendicular to current
 - D. No relation with current

14. If charge $Q = 96500 \text{ C}$, number of electrons is approximately:

- A. 6×10^{23}
- B. 1.6×10^{-19}
- C. 6.02×10^{23}
- D. 3.2×10^6

15. **Steady-state current means:**

- A. Zero current
- B. Current constant with time
- C. Current increasing with time
- D. Maximum current

 End of partial Section 1

Reply “Continue” and I will render the next part (Q16 onward).

continue

Continuing your **line-by-line rendered Markdown** from where we left off:

SECTION 1: ELECTRIC CURRENT & CHARGE FLOW (Q1–30) — Continued

16. **Current density J and electric field E are related as:**

- A. $J = E/\sigma$
- B. $J = \sigma E$
- C. $J = E \cdot \rho$
- D. $J = E/\rho$

17. **In a cross-section, if area increases and current remains same:**

- A. Current density increases
- B. Current density decreases
- C. Current density unchanged
- D. Cannot be determined

18. **If electrons move with average drift velocity v_o :**

- A. $I = nAev_o$

B. $I = nAv_0$

C. $I = nAe$

D. $I = Av_0$

19. Free electron density in metals is approximately:

A. 10^{23} m^{-3}

B. 10^{28} m^{-3}

C. 10^{15} m^{-3}

D. 10^{30} m^{-3}

20. Current through cross-section depends on:

A. Length of conductor only

B. Cross-sectional area only

C. Applied voltage and properties

D. Material only

21. In conductors, charge carriers are:

A. Ions

B. Holes

C. Free electrons

D. Protons

22. Conservation of charge means:

A. Charge cannot be created or destroyed

B. Charge can be created

C. Charge only in metals

D. Charge static always

23. If $I = 0.5 \text{ A}$ and $t = 4 \text{ s}$, charge is:

A. 0.125 C

B. 0.5 C

C. 2 C

D. 8 C

24. Drift velocity in a metal is typically:

A. 10^{-4} m/s

B. 10^5 m/s

C. 10^2 m/s

D. 10^8 m/s

25. Which statement is correct?

A. All moving charges constitute current

B. Only electron motion matters

- C. Conventional current = actual flow
D. Direction arbitrary
26. **In a series circuit, current:**
A. Varies in different sections
B. Same everywhere
C. Zero at some points
D. Unpredictable
27. **Temperature increase affects electron motion by:**
A. Increasing drift velocity
B. Increasing thermal velocity
C. Decreasing conductivity
D. Both B and C
28. **If 1.6×10^{19} electrons pass, charge is:**
A. 0.1 C
B. 1 C
C. 10 C
D. 100 C
29. **Current becomes zero when:**
A. Potential difference is zero
B. Charge stops flowing
C. Circuit is complete
D. Both A and B
30. **Number of electrons per second through area for $I = 1$ A:**
A. 6.25×10^{18}
B. 1.6×10^{19}
C. 3.2×10^{18}
D. 9.1×10^{18}
-

 **Section 1 complete**

Reply "Continue" and I will start **SECTION 2: Drift Velocity & Mobility (Q31–60)**.

SECTION 2: DRIFT VELOCITY & MOBILITY (Q31–60)

31. **Drift velocity is:**
- A. Same as thermal velocity
 - B. Much less than thermal velocity
 - C. Equal to thermal velocity
 - D. Random motion only
32. **Relation between drift velocity and current is:**
- A. $v_d = I/(nAe)$
 - B. $v_d = nAeI$
 - C. $v_d = I \cdot n \cdot A \cdot e$
 - D. $v_d = n/(IAe)$
33. **If current doubles, drift velocity:**
- A. Doubles
 - B. Halves
 - C. Remains same
 - D. Becomes zero
34. **Mobility is defined as:**
- A. Velocity per unit field
 - B. Field per unit velocity
 - C. Current per unit field
 - D. Charge per unit field
35. **SI unit of mobility is:**
- A. $\text{m}^2/(\text{V} \cdot \text{s})$
 - B. $\text{V} \cdot \text{s}/\text{m}^2$
 - C. m/s
 - D. V/s
36. **Mobility depends on:**
- A. Temperature only
 - B. Material only
 - C. Material and temperature
 - D. Applied voltage
37. **If electric field doubles, drift velocity:**
- A. Doubles

- B. Halves
- C. Remains same
- D. Becomes zero

38. $\mu = vd/E$ where E is:

- A. Potential
- B. Electric field
- C. Potential gradient
- D. Both B and C

39. Relaxation time τ relates to:

- A. Free fall time between collisions
- B. Total motion time
- C. Time period of oscillation
- D. Half-life of charge

40. Mobility and relaxation time are related as:

- A. $\mu = e\tau/m$
- B. $\mu = m\tau/e$
- C. $\mu = \tau/m$
- D. $\mu = m/(e\tau)$

41. Drift velocity in a copper wire ($I = 10$ A, $A = 1$ mm²) is approximately:

- A. 10^{-4} m/s
- B. 10^{-2} m/s
- C. 10 m/s
- D. 10^2 m/s

42. If area of cross-section increases, drift velocity:

- A. Increases
- B. Decreases
- C. Remains same
- D. Becomes undefined

43. Thermal velocity of electrons is:

- A. 10^2 m/s
- B. 10^4 m/s
- C. 10^6 m/s
- D. 10^8 m/s

44. Drift velocity increases when:

- A. Electric field increases
- B. Temperature increases

C. Pressure increases

D. None of above

45. **Mobility of electrons in semiconductors is:**

A. Greater than metals

B. Less than metals

C. Equal to metals

D. Undefined

46. **Free path of electrons is:**

A. Distance between collisions

B. Total path length

C. Radius of conductor

D. Wavelength of electron

47. **Mean free path increases with:**

A. Temperature increase

B. Pressure decrease

C. Field increase

D. Impurity increase

48. **If $v_d = 10^{-5}$ m/s and $E = 10$ V/m, mobility is:**

A. 10^{-6} m²/(V·s)

B. 10^{-5} m²/(V·s)

C. 10^{-4} m²/(V·s)

D. 10^{-3} m²/(V·s)

49. **In a metal, electrons:**

A. Flow in one direction continuously

B. Move randomly with slight drift

C. Stationary

D. Move in circles

50. **Current $I = nAe\mu E$ where n is:**

A. Total electrons

B. Number density

C. Electron count

D. Charge number

51. **Drift velocity formula $I = nAev_d$ gives:**

A. Instantaneous current

B. Average current

C. Maximum current

D. Minimum current

52. **If temperature increases, thermal velocity:**

A. Increases

B. Decreases

C. Remains constant

D. Becomes zero

53. **Number density n for copper is:**

A. 10^{23} m^{-3}

B. 10^{28} m^{-3}

C. 10^{33} m^{-3}

D. 10^{18} m^{-3}

54. **If $v_d = I/(nAe)$ and I increases, v_d :**

A. Increases proportionally

B. Decreases

C. Remains constant

D. Becomes nonlinear

55. **Mobility is inversely proportional to:**

A. Temperature

B. Number density

C. Applied field

D. Cross-section

56. **Relaxation time represents:**

A. Collision frequency

B. Time between collisions

C. Acceleration time

D. Deceleration time

57. **In a thicker wire, drift velocity for the same current:**

A. Increases

B. Decreases

C. Remains same

D. Undefined

58. **Mobility formula $\mu = e\tau/m$ shows:**

A. μ depends on charge and mass

B. μ inversely proportional to mass

C. μ depends on relaxation time

D. All of above

59. If $I = 5 \text{ A}$, $n = 10^{28} \text{ m}^{-3}$, $A = 2 \text{ mm}^2$, v_d is:

A. $1.56 \times 10^{-5} \text{ m/s}$

B. $1.56 \times 10^{-4} \text{ m/s}$

C. $1.56 \times 10^{-3} \text{ m/s}$

D. $1.56 \times 10^{-2} \text{ m/s}$

60. Drift velocity and thermal velocity are:

A. In same direction

B. In opposite directions

C. Perpendicular

D. Unrelated in direction

☒ Section 2 complete

Reply "Continue" when ready for SECTION 3: Resistance & Resistivity (Q61–100).

continue

SECTION 3: RESISTANCE & RESISTIVITY (Q61–100)

61. Resistance is defined as:

A. V/I

B. I/V

C. $V \cdot I$

D. I/V^2

62. SI unit of resistance is:

A. Ampere

B. Ohm

C. Volt

D. Coulomb

63. Resistivity formula is:

A. $\rho = RA/L$

B. $\rho = RL/A$

C. $\rho = A/(RL)$

D. $\rho = L/(RA)$

64. **SI unit of resistivity is:**

A. Ω

B. Ω/m

C. $\Omega \cdot \text{m}$

D. S/m

65. **Resistance depends on:**

A. Length only

B. Area only

C. Material only

D. All of above

66. **If length doubles, resistance:**

A. Doubles

B. Halves

C. Remains same

D. Quadruples

67. **If cross-sectional area doubles, resistance:**

A. Doubles

B. Halves

C. Remains same

D. Quadruples

68. **Resistivity of copper at room temperature is approximately:**

A. $1.7 \times 10^{-8} \Omega \cdot \text{m}$

B. $1.7 \times 10^{-6} \Omega \cdot \text{m}$

C. $1.7 \times 10^{-4} \Omega \cdot \text{m}$

D. $1.7 \times 10^{-2} \Omega \cdot \text{m}$

69. **Conductivity σ is related to resistivity as:**

A. $\sigma = \rho$

B. $\sigma = 1/\rho$

C. $\sigma = \rho^2$

D. $\sigma = \sqrt{\rho}$

70. **If wire is stretched to double length:**

A. Resistance doubles

B. Resistance halves

C. Resistance becomes 4 times

D. Resistance unchanged

71. **When wire is cut into 2 equal parts:**

A. Both have same resistance

B. Total resistance doubles

C. Each part has half resistance

D. Resistance unchanged

72. **Temperature coefficient of resistance α means:**

A. R increases by α per $^{\circ}\text{C}$

B. R increases by $\alpha\%$ per $^{\circ}\text{C}$

C. R increases by αR_0 per $^{\circ}\text{C}$

D. R increases by $\alpha/^{\circ}\text{C}$

73. **For metals, temperature coefficient is:**

A. Positive

B. Negative

C. Zero

D. Variable

74. **For semiconductors, temperature coefficient is:**

A. Positive

B. Negative

C. Zero

D. Infinite

75. **Ohm's law states:**

A. $I = V \cdot R$

B. $V = I/R$

C. $V = IR$

D. $R = I/V$

76. **Ohm's law is valid for:**

A. All materials

B. Ohmic materials only

C. Non-ohmic devices

D. Semiconductors

77. **Non-ohmic devices include:**

A. Copper wire

B. Nichrome wire

- C. Diode
- D. Resistor

78. **V-I graph for ohmic resistor is:**

- A. Straight line
- B. Curved
- C. Parabola
- D. Hyperbola

79. **V-I graph for non-ohmic device is:**

- A. Straight line
- B. Curved
- C. Linear
- D. Horizontal

80. **Conductance G is:**

- A. $R \cdot I$
- B. V/I
- C. $1/R$
- D. R/V

81. **SI unit of conductance is:**

- A. Ohm
- B. Siemens
- C. Ampere
- D. Volt

82. **Specific resistance another name for:**

- A. Conductance
- B. Resistance
- C. Resistivity
- D. Conductivity

83. **If $R = 10 \Omega$, $L = 2 \text{ m}$, $A = 1 \text{ mm}^2$, ρ is:**

- A. $5 \times 10^{-6} \Omega \cdot \text{m}$
- B. $5 \times 10^{-5} \Omega \cdot \text{m}$
- C. $5 \times 10^{-4} \Omega \cdot \text{m}$
- D. $5 \times 10^{-3} \Omega \cdot \text{m}$

84. **Nichrome wire used because:**

- A. Low resistance
- B. High melting point

- C. Low cost
- D. Easy to bend

85. **Copper preferred for wiring because:**

- A. High resistance
- B. Low resistivity
- C. Low melting point
- D. Easy to oxidize

86. $R_0 = 100\ \Omega$ at 0°C , $\alpha = 0.005/^\circ\text{C}$. At 100°C , R is:

- A. $105\ \Omega$
- B. $150\ \Omega$
- C. $200\ \Omega$
- D. $50\ \Omega$

87. **If two resistors R_1 and R_2 are in series:**

- A. $R = R_1 + R_2$
- B. $1/R = 1/R_1 + 1/R_2$
- C. $R = R_1 \cdot R_2$
- D. $R = (R_1 + R_2)/(R_1 \cdot R_2)$

88. **If two resistors are in parallel:**

- A. $R = R_1 + R_2$
- B. $1/R = 1/R_1 + 1/R_2$
- C. $R = R_1/R_2$
- D. $R = \sqrt{R_1 \cdot R_2}$

89. **Three $2\ \Omega$ resistors in series give:**

- A. $2/3\ \Omega$
- B. $2\ \Omega$
- C. $6\ \Omega$
- D. $8\ \Omega$

90. **Three $2\ \Omega$ resistors in parallel give:**

- A. $2/3\ \Omega$
- B. $2\ \Omega$
- C. $6\ \Omega$
- D. $8\ \Omega$

91. **Resistance increases with:**

- A. Length increase
- B. Area increase

- C. Temperature decrease
- D. Impurity decrease

92. In a heating wire, greater heat is produced in:

- A. Thicker part
- B. Thinner part
- C. Both equal
- D. Depends on length

93. Macroscopic Ohm's law is:

- A. $V = IR$
- B. $J = \sigma E$
- C. $E = \rho J$
- D. Both B and C

94. For copper, ρ increases as:

- A. Temperature increases
- B. Temperature decreases
- C. Pressure increases
- D. Area changes

95. Wire of resistance R is cut into 4 equal parts. Resistance of each part:

- A. $R/4$
- B. $R/2$
- C. R
- D. $4R$

96. When all 4 parts are connected in parallel:

- A. $R/16$
- B. $R/4$
- C. R
- D. $4R$

97. Conductivity formula is:

- A. $\sigma = 1/\rho$
- B. $\sigma = \rho$
- C. $\sigma = \rho/L$
- D. $\sigma = L/\rho$

98. Conductivity unit is:

- A. $\Omega \cdot m$
- B. $(\Omega \cdot m)^{-1}$

C. Ω

D. A/V

99. **Material with lowest resistivity:**

A. Silver

B. Copper

C. Aluminum

D. Gold

100. **Material with highest resistivity (insulator):**

A. Glass

B. Mica

C. Rubber

D. All have similar high resistivity

☒ **Section 3 complete**

Reply “Continue” to proceed with **SECTION 4: EMF & Internal Resistance (Q101–130)**.

continue

SECTION 4: EMF & INTERNAL RESISTANCE (Q101–130)

101. **EMF is defined as:**

A. Potential difference across terminals

B. Energy per unit charge

C. Force on charge

D. Work done by force

102. **SI unit of EMF is:**

A. Joule

B. Watt

C. Volt

D. Ampere

103. **Internal resistance is:**

A. Resistance of wire

B. Resistance of battery itself

C. Resistance of load

D. Net resistance

104. **Terminal voltage formula is:**

A. $V = \varepsilon + Ir$

B. $V = \varepsilon - Ir$

C. $V = \varepsilon/Ir$

D. $V = \varepsilon Ir$

105. **When battery is discharging:**

A. $V > \varepsilon$

B. $V < \varepsilon$

C. $V = \varepsilon$

D. $V = 0$

106. **When battery is charging:**

A. $V > \varepsilon$

B. $V < \varepsilon$

C. $V = \varepsilon$

D. $V = Ir$

107. **Ideal battery has:**

A. High internal resistance

B. Zero internal resistance

C. Variable internal resistance

D. Infinite resistance

108. **Real battery has:**

A. Zero internal resistance

B. Non-zero internal resistance

C. Infinite resistance

D. No EMF

109. **Current from battery is:**

A. $I = V/(R + r)$

B. $I = \varepsilon/(R + r)$

C. $I = \varepsilon R/r$

D. $I = \varepsilon/(R - r)$

110. **If internal resistance increases:**

A. Current increases

B. Terminal voltage increases

C. Both increase

D. Both decrease

111. $\epsilon = 12 \text{ V}$, $r = 1 \Omega$, $R = 11 \Omega$. Current is:

A. 0.5 A

B. 1 A

C. 1.2 A

D. 12 A

112. From Q111, terminal voltage is:

A. 11 V

B. 12 V

C. 13 V

D. 10 V

113. Short circuit condition when:

A. $R = 0$

B. $\epsilon = 0$

C. $r = 0$

D. $V = \epsilon$

114. In short circuit, current:

A. $I = \epsilon/r$

B. $I = V/R$

C. $I = 0$

D. Maximum

115. Terminal voltage becomes zero when:

A. $R = 0$

B. $\epsilon = 0$

C. $I = 0$

D. $r = \infty$

116. Maximum power delivered to external circuit when:

A. $R \gg r$

B. $R \ll r$

C. $R = r$

D. $R = 2r$

117. Power dissipated in internal resistance is:

A. ϵI

B. VI

C. $I^2 r$

D. V^2/R

118. Power supplied by source is:

A. VI

B. ϵI

C. $I^2 R$

D. V^2/r

119. Efficiency of battery is:

A. ϵ/V

B. V/ϵ

C. $R/(R + r)$

D. $r/(R + r)$

120. Two cells in series with EMF ϵ_1 and ϵ_2 :

A. $\epsilon = \epsilon_1 + \epsilon_2$

B. $\epsilon = \epsilon_1 - \epsilon_2$

C. $\epsilon = \epsilon_1 \cdot \epsilon_2$

D. $\epsilon = \sqrt{(\epsilon_1 \cdot \epsilon_2)}$

121. Two cells in parallel (same ϵ):

A. $\epsilon = 2\epsilon_1$

B. $\epsilon = \epsilon_1/2$

C. $\epsilon = \epsilon_1$

D. $\epsilon = 0$

122. Internal resistance in series connection:

A. $r = r_1 + r_2$

B. $r = r_1/r_2$

C. $r = (r_1 + r_2)/2$

D. $1/r = 1/r_1 + 1/r_2$

123. Internal resistance in parallel connection:

A. $r = r_1 + r_2$

B. $r = r_1/r_2$

C. $1/r = 1/r_1 + 1/r_2$

D. $r = (r_1 + r_2)/(r_1 r_2)$

124. Back EMF in charging:

A. Opposes applied voltage

B. Aids applied voltage

C. Equal to applied voltage

D. Zero during charging

125. Which statement is true?

A. EMF = Terminal voltage always

B. EMF > Terminal voltage (discharging)

C. EMF < Terminal voltage (charging)

D. Both B and C

126. Heat generated in cell due to:

A. I^2R only

B. I^2r only

C. $I^2(R + r)$

D. $IR + Ir$

127. When battery supplies maximum power to load:

A. $R = 0$

B. $R = r$

C. $R = 2r$

D. $R = \infty$

128. Efficiency is maximum when:

A. $R \gg r$

B. $R = r$

C. $R \ll r$

D. $R = 2r$

129. Energy stored in battery:

A. ϵQ

B. I^2Rt

C. V^2/R

D. Pt

130. Deep discharge means:

A. Complete discharge

B. Partial discharge

C. No discharge

D. Overcharge

continue

SECTION 5: OHM'S LAW & KIRCHHOFF'S LAWS (Q131–170)

131. Kirchhoff's first law (junction rule) states:

- A. $\Sigma V = 0$
- B. $\Sigma I = 0$
- C. $\Sigma I_{\text{in}} = \Sigma I_{\text{out}}$
- D. $\Sigma R = \text{constant}$

132. Kirchhoff's second law (loop rule) states:

- A. $\Sigma V = 0$
- B. $\Sigma I = 0$
- C. $\Sigma P = 0$
- D. $\Sigma R = 0$

133. In a series circuit, by Kirchhoff's first law:

- A. Currents different
- B. Currents same
- C. Voltages same
- D. Resistances same

134. In a parallel circuit, by Kirchhoff's first law:

- A. Current same
- B. Current divides
- C. Voltage same
- D. Resistance same

135. In a closed loop, sum of all potential differences:

- A. Positive
- B. Negative
- C. Zero
- D. Maximum

136. Kirchhoff's laws based on:

- A. Conservation of charge
- B. Conservation of energy

- C. Both A and B
- D. Conservation of mass

137. **In a circuit with 2 loops, independent equations:**

- A. 1
- B. 2
- C. 3
- D. 4

138. **To solve multi-loop circuits, we use:**

- A. Ohm's law only
- B. Kirchhoff's laws only
- C. Both Ohm and Kirchhoff
- D. None

139. **Kirchhoff's voltage law applies to:**

- A. DC circuits only
- B. AC circuits only
- C. Both DC and AC
- D. No circuits

140. **Current conservation means:**

- A. Current increases
- B. Current decreases
- C. Current entering = current leaving
- D. No current flows

141. **In circuit analysis, choosing loop direction:**

- A. Must be clockwise
- B. Can be arbitrary
- C. Must be counterclockwise
- D. No direction matters

142. **Sign convention for EMF in loop:**

- A. Always positive
- B. Always negative
- C. Positive if aids loop direction
- D. Negative if opposing

143. **Sign convention for potential drops:**

- A. Negative when current aids loop
- B. Positive when current aids loop

- C. Zero always
- D. Depends on resistance

144. In a parallel circuit, applying junction rule:

- A. $I = I_1 + I_2 + I_3 \dots$
- B. $V = V_1 + V_2 \dots$
- C. $R = R_1 + R_2 \dots$
- D. None

145. Equivalent resistance of series resistors:

- A. $1/R = 1/R_1 + 1/R_2$
- B. $R = R_1 + R_2$
- C. $R = R_1 \cdot R_2 / (R_1 + R_2)$
- D. $R = \sqrt{R_1 \cdot R_2}$

146. Equivalent resistance of parallel resistors:

- A. $R = R_1 + R_2$
- B. $1/R = 1/R_1 + 1/R_2$
- C. $R = R_1 \cdot R_2$
- D. $R = R_1 - R_2$

147. Nodal voltage method uses:

- A. Loop rule
- B. Current law
- C. Voltage law
- D. None

148. Mesh current method uses:

- A. Current law
- B. Voltage law
- C. Power law
- D. None

149. Thevenin's theorem application:

- A. Simplifies circuits
- B. Finds equivalent resistance
- C. Calculates maximum power
- D. All of above

150. Norton's theorem is equivalent to:

- A. Thevenin theorem
- B. Superposition theorem

C. Different from Thevenin

D. Only for AC

151. **Superposition principle states:**

A. All sources act simultaneously

B. Each source acts independently

C. Combine individual effects

D. Both B and C

152. **In a series circuit with Kirchhoff's law:**

A. $\varepsilon_1 + \varepsilon_2 = I(R_1 + R_2 + r_1 + r_2)$

B. $\varepsilon_1 - \varepsilon_2 = I(R_1 + R_2)$

C. $\varepsilon_1 = IR_1$

D. Depends on polarity

153. **Maximum number of independent loop equations:**

A. Equal to branches

B. Equal to loops

C. $l = b - n + 1$

D. $l = n - b$

154. **Where l = loops, b = branches, n = nodes:**

A. $b = l + n$

B. $b = l + n - 1$

C. $b = n + l$

D. $b = l$

155. **Using Kirchhoff's laws is:**

A. Tedious for complex circuits

B. Always necessary

C. Not needed with Ohm's law

D. Sufficient alone

156. **Path independence means:**

A. Work done depends on path

B. Work done independent of path

C. No work done

D. Variable work

157. **Potential difference in a circuit:**

A. Same everywhere

B. Depends on path

C. Path independent

D. Variable

158. **Loop equations are:**

A. Dependent always

B. Linearly independent

C. Both types

D. Indeterminate

159. **Null measurement principle uses:**

A. Galvanometer

B. Zero current condition

C. Potentiometer

D. All above

160. **Potentiometer measures:**

A. Current accurately

B. Voltage with no current

C. Resistance directly

D. Power

161. **Meter bridge balancing gives:**

A. $P/Q = R/S$

B. $P/Q = S/R$

C. $P/R = Q/S$

D. $P \cdot Q = R \cdot S$

162. **In meter bridge, null deflection when:**

A. Current flows through galvanometer

B. No current through galvanometer

C. Resistance equal

D. Voltage equal

163. **Wheatstone bridge formula:**

A. $P \cdot S = Q \cdot R$

B. $P/Q = R/S$

C. $P \cdot R = Q \cdot S$

D. $P + R = Q + S$

164. **Sensitivity of bridge:**

A. Independent of resistance

B. Depends on balance point

C. Maximum at center

D. Zero at ends

165. **Application of Kirchhoff's law to AC:**

A. Not applicable

B. Applicable with modifications

C. Same as DC

D. Not proven

166. **Charge conservation in circuit:**

A. By Kirchhoff's current law

B. By Kirchhoff's voltage law

C. By Ohm's law

D. By power law

167. **Network analysis requires:**

A. Ohm's law only

B. Kirchhoff's laws only

C. Both and additional theorems

D. No laws

168. **Mesh analysis suitable for:**

A. Series circuits only

B. Parallel circuits only

C. Multi-loop circuits

D. Single loop

169. **Number of unknown currents equals:**

A. Number of nodes

B. Number of loops

C. Number of branches

D. Number of elements

170. **System of equations for circuit:**

A. Always unique solution

B. May have no solution

C. May have multiple solutions

D. Both B and C

Reply "Continue" for the final part: **SECTION 6: Series & Parallel Combinations (Q171–200)**.

Continue"

SECTION 6: SERIES & PARALLEL COMBINATIONS (Q171–200)

171. **Series connection characteristics:**

- A. Same current
- B. Same voltage
- C. Same resistance
- D. Same power

172. **Parallel connection characteristics:**

- A. Same current
- B. Different current
- C. Same voltage
- D. Same power

173. **Two equal resistors in series:**

- A. $R_{eq} = R$
- B. $R_{eq} = 2R$
- C. $R_{eq} = R/2$
- D. $R_{eq} = R^2$

174. **Two equal resistors in parallel:**

- A. $R_{eq} = R$
- B. $R_{eq} = 2R$
- C. $R_{eq} = R/2$
- D. $R_{eq} = R^2$

175. **Current divider rule for R_1 and R_2 in parallel:**

- A. $I_1 = I(R_1/R_2)$
- B. $I_1 = I \cdot R_2 / (R_1 + R_2)$
- C. $I_1 = I \cdot R_1 / (R_1 + R_2)$
- D. $I_1 = I / (R_1 + R_2)$

176. **Voltage divider rule for R_1 and R_2 in series:**

- A. $V_1 = V \cdot R_1 / (R_1 + R_2)$
- B. $V_1 = V \cdot R_2 / (R_1 + R_2)$

C. $V_1 = V \cdot (R_1 + R_2) / R_1$

D. $V_1 = V \cdot R_2 / R_1$

177. **Three 6Ω resistors in series:**

A. $2\ \Omega$

B. $6\ \Omega$

C. $18\ \Omega$

D. $36\ \Omega$

178. **Three 6Ω resistors in parallel:**

A. $2\ \Omega$

B. $6\ \Omega$

C. $18\ \Omega$

D. $36\ \Omega$

179. **2Ω and 3Ω in series:**

A. $0.4\ \Omega$

B. $1.2\ \Omega$

C. $5\ \Omega$

D. $6\ \Omega$

180. **2Ω and 3Ω in parallel:**

A. $0.4\ \Omega$

B. $1.2\ \Omega$

C. $5\ \Omega$

D. $6\ \Omega$

181. **Series circuit advantage:**

A. Simple voltage distribution

B. Easy to open/close

C. High reliability

D. Equal current everywhere

182. **Parallel circuit advantage:**

A. Equal voltage all branches

B. Low total resistance

C. Can add/remove branches

D. All of above

183. **Series circuit disadvantage:**

A. If one breaks, all fail

B. Simple connections

C. Easy to troubleshoot

D. Low cost

184. **Parallel circuit disadvantage:**

A. Complex wiring

B. High current from source

C. Bulky connections

D. All of above

185. **Voltage across each resistor in series:**

A. $V = V_{\text{total}}/n$

B. $V = V_{\text{total}}$ (each)

C. $V \propto \text{Resistance}$

D. Both A and C

186. **Current through each resistor in parallel:**

A. $I = I_{\text{total}}/n$

B. $I = I_{\text{total}}$ (each)

C. $I \propto 1/\text{Resistance}$

D. Both A and C

187. **If $R_1 \gg R_2$ in series:**

A. Most voltage across R_1

B. Most voltage across R_2

C. Equal voltage

D. Voltage zero

188. **If $R_1 \gg R_2$ in parallel:**

A. Most current through R_1

B. Most current through R_2

C. Equal current

D. Current zero

189. **Power dissipation in series:**

A. $P \propto R$ (each resistor)

B. $P = \text{same}$ (all)

C. $P = I^2 R$

D. $P \propto V$

190. **Power dissipation in parallel:**

A. $P \propto R$

B. $P \propto 1/R$

C. $P = \text{same (all)}$

D. $P = V^2/R$

191. **In a mixed circuit with series-parallel:**

A. Use series rules throughout

B. Use parallel rules throughout

C. Identify sections, apply appropriate rules

D. Cannot solve

192. **Equivalent resistance of n identical resistors in series:**

A. $R_{eq} = R/n$

B. $R_{eq} = nR$

C. $R_{eq} = R$

D. $R_{eq} = R^2$

193. **Equivalent resistance of n identical resistors in parallel:**

A. $R_{eq} = R/n$

B. $R_{eq} = nR$

C. $R_{eq} = R$

D. $R_{eq} = n/R$

194. **Bridge circuit is balanced when:**

A. Current flows

B. $P/Q = R/S$

C. All resistances equal

D. Current zero

195. **Y- Δ transformation used for:**

A. Simplifying non-planar networks

B. Converting 3-resistor connections

C. Equivalent circuit analysis

D. All of above

196. **Ladder network solved by:**

A. Series-parallel reduction

B. Successive simplification

C. Mathematical recursion

D. All methods

197. **Maximum power transfer when:**

A. $R \gg r$

B. $R = 0$

C. $R = r_{Th}$

D. $R = \infty$

198. In external circuit $R =$ internal resistance:

A. Current maximum

B. Power maximum

C. Voltage maximum

D. Efficiency maximum

199. Efficiency in maximum power transfer:

A. 25%

B. 50%

C. 75%

D. 100%

200. Load resistance for maximum power:

A. Zero

B. Infinity

C. Equal to source resistance

D. Twice source resistance